Answer **all** the questions in the spaces provided.

1 Two spheres A and B of masses 3.7 kg and 2.7 kg respectively are attached to two light inextensible strings L_1 and L_2 each of length 0.50 m.

The combination of masses is made to swing in a vertical circle with both strings always in a straight line, as shown in Fig. 1.1.

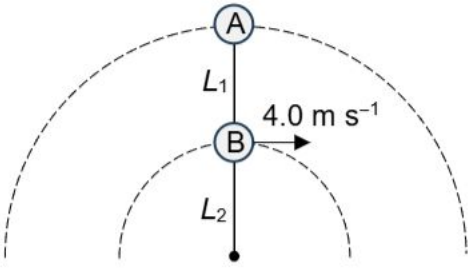


Fig 1.1

At the top of its motion, mass B is travelling at 4.0 m s^{-1}.

(a) Determine the speed of mass A at this instant.

Both masses have the same angular velocity.

$v = r\omega$

$\frac{v_A}{r_A} = \frac{v_B}{r_B} \Rightarrow \frac{v_A}{0.5+0.5} = \frac{4.0}{0.5}$ [M1]

$v_A = 8.0$ m s^{-1} [A1]

speed of mass A = ... m s^{-1} [2]

(b) Show that the tension in L_1 at this instant is 200 N.

$W_A + T_A = \frac{mv^2}{r}$

$T_A = \frac{mv^2}{r} - W_A$

$= \frac{(3.7)(8.0)^2}{0.50+0.50} - (3.7)(9.81)$ [M1]

$= 200.5 = 200$ N [A0]

[1]

(c) Hence or otherwise, determine the tension in L_2 at this instant.

$$W_B + T_B - T_A = \frac{m_B v_B^{\,2}}{r_B}$$

$$T_B = \frac{m_B v_B^{\,2}}{r_B} - W_B + T_A$$

$$= \frac{(2.7)(4.0)^2}{0.50} - (2.7)(9.81) + 201 \qquad \text{[M1]}$$

$$= 261 \text{ N} \qquad \text{[A1]}$$

tension in L_2 = ... N [2]

(d) At another instant in time, the combination of masses is at the bottom of its motion as shown in Fig. 1.2.

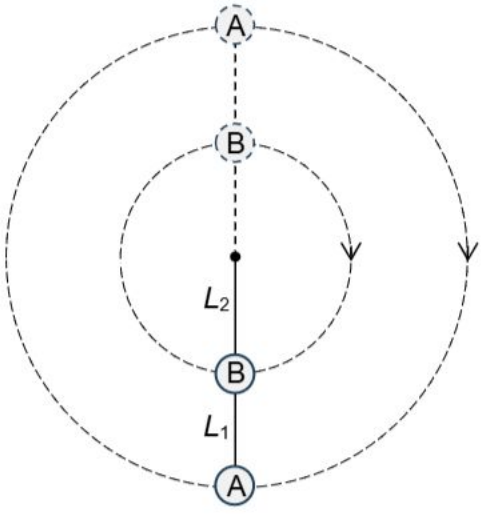


Fig 1.2

A student suggests that because the tensions in strings L_1 and L_2 change continuously during the motion, the total mechanical energy (i.e. kinetic and potential energy) of the system is not conserved.

With reference to tensions in the strings, state whether you agree with this statement and explain your reasoning.

I disagree with the statement, as tension is **always** acting perpendicular to the direction of motion of both masses [M1] Hence there is no work done by tension and hence total mechanical energy of the system is conserved. [A1]

I disagree with the statement, as tension is **always** acting perpendicular to the direction of motion of both masses. [M1] Hence there is no work done by tension but there may be other resistive forces and hence mechanical energy is not conserved. [A1]

[2]

[Total: 7]